

M. BILAL JAVED

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Location: Islamabad

SUMMARY

Final-semester Computer Science student specializing in AI/ML, Generative AI, and full-stack development. Experienced in building scalable AI systems from model development to production deployment. Skilled in transformer-based NLP, computer vision, multi-agent LLM workflows, and cloud-native application development. Passionate about applying AI to real-world automation, accessibility, and analytics problems.

EDUCATION & Experience

Junior AI Engineer - Annovasol

Dec 2025 - Apr 2026

- Designed and deployed 3+ GenAI-powered automation workflows using n8n, integrating Gemini and OpenAI LLM APIs to automate business processes, reducing manual effort by an estimated 40%.
- Built end-to-end AI pipelines including prompt engineering, API orchestration, data validation, and structured output handling.

AI intern - VisionX

June 2025 - Aug 2025

- Developed a YOLO + X3D computer vision pipeline for retail shoplifting detection, achieving 85%+ accuracy on a custom dataset of 2,000+ annotated frames, with a ~15% precision gain over pretrained baselines.
- Implemented a facial recognition module integrated into the detection system, enabling real-time security alerts for retail store operators.

Bachelors in Computer Science - FAST NUCES

Aug 2022 - June 2026

- Specializing in AI/ML and Software development.

PROJECTS

Classroom Student Engagement Analysis (Final-Year Project)

Sep 2025 - Apr 2026

- Built a computer vision pipeline using YOLO and a custom emotion-detection CNN to measure student engagement from lecture videos, achieving 80%+ accuracy across 10+ hours of test footage.
- Designed FastAPI backend and Power BI dashboards to store and present per-student engagement metrics, demonstrating a full AI-to-deployment workflow.

Multi-Agent Prescription-to-Audio Workflow (Urdu)

Nov 2025 - Dec 2025

- Created a multi-agent AI workflow where an LLM (Gemini API) extracts and validates data from prescription images and generates simplified explanations.
- Converted the LLM output into Urdu audio using Google TTS to improve accessibility for elderly patients, and tested on 50+ real prescriptions.

Platform for Semantic Analysis of YouTube Comments

Oct 2025 - Nov 2025

- Developed a full-stack platform (FastAPI + PostgreSQL + React) performing sentiment and semantic analysis via HuggingFace BERT/roBERTa on 10,000+ comments per run with an interactive results dashboard.
- Visualized processed insights through an interactive UI, showcasing end-to-end integration of AI pipelines with web technologies.

Shoplifting Detection System

Jul 2025 - Jul 2025

- Built a CV pipeline using YOLO for detection and X3D for action recognition to identify suspicious behavior in retail environments.
- Fine-tuned models on custom datasets to improve shoplifting detection accuracy and support real-time security alerts.

TECHNICAL SKILLS:

- Languages: Python, Java, C/C++, Kotlin, Flutter
- Cloud(DevSecOps): AWS (EC2, Lambda, RDS, EKS, SageMaker), Docker, Kubernetes, Git, GitHub
- AI/ML: LLM, LangChain, LangSmith, Tensorflow, PyTorch, Transformers, GANs, HuggingFace, RAG, n8n
- Web&Data: FastAPI, RESTful APIs, PostgreSQL, MySQL, Power BI, Matplotlib, Seaborn

CERTIFICATIONS & ACHIEVEMENTS:

- AWS Cloud Foundation Course (Simplilearn & AWS Educate)
- Deep Learning Specialization by AndrewNg (Coursera)
- Awarded a scholarship for outstanding academic performance in FSC from PEEF